IntellizenHire

Aabha Bhide Aditi Gadhave Niharika Bandekar Sumeet Gupta
Guide : Prof. Vijaya Bharathi Jagan

NLP-Based Resume Analysis and Ranking System

INTRODUCTION

- Traditional resume screening and keyword-based ATS systems fail to capture semantic relevance and to provide useful feedback.
- The proposed platform automates resume analysis and job matching using a weighted ATS score.
- It also provides skill gap detection, recommendations, mock interviews and explainable candidate ranking.

UNIQUENESS

- Dual student-HR platform with explainable candidate ranking.
- Semantic NLP-based resume-job matching beyond keyword ATS systems.
- Interpretable weighted ATS scoring with skill gap detection and recommendations.

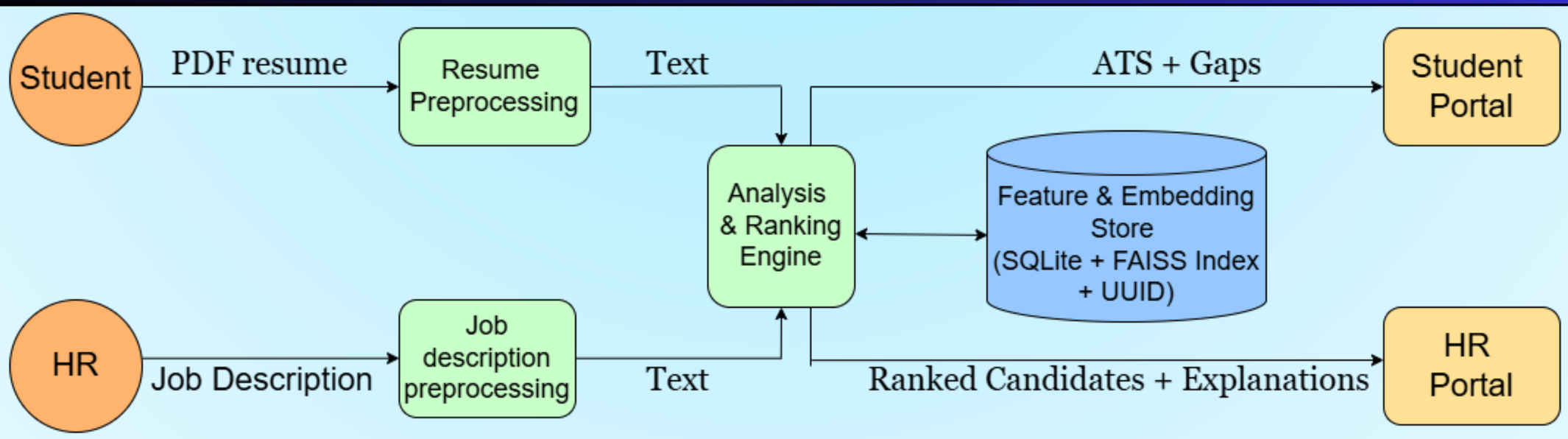
OBJECTIVES

- To compute an interpretable ATS score for resume–job compatibility.
- To detect skill gaps and provide feedback to improve resumes.
- To support HR in managing job postings, applications and candidate ranking using semantic similarity.
- To ensure transparent and reproducible evaluation.

METHODOLOGY

- ATS Score: Weighted evaluation using keywords, formatting, experience similarity, skills and education.
- Skill Gap Detection: Identifies missing skills by comparing resume and job requirements.
- Candidate Ranking: Ranks candidates using ATS score and extracted features.
- Explainability: Provides explanations for ranking results.

SYSTEM ARCHITECTURE



RESULTS

Student Portal

- ATS score with resume insights and skill gap detection.
- Job recommendations, mock interview questions and progress tracking.

HR Portal

- Bulk resume analysis with ATS scoring.
- Candidate ranking and application management.

CONCLUSION & FUTURE WORK

- Provides an automated and transparent recruitment support system that helps students improve resumes and assists recruiters in candidate selection.
- Enhances recruitment efficiency through interpretable analytics and explainable ranking.
- Future Work: Integrate additional data sources (assessments, portfolios, interview feedback) and improve scoring with adaptive, bias-aware evaluation.